Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the boxes is helpful for understanding the concept(s).

16.1 Nominal GDP and Real GDP

- Meanings of nominal GDP and real GDP

Nominal GDP	Real GDP
• It refers to the GDP measured in terms of market prices in the current year. • The nominal GDP of a particular year is calculated by multiplying the amount of output of that year by the prices in that year. • In Hong Kong, the Census and the Statistics Department uses GDP at current market prices to reflect nominal GDP.	• It refers to the GDP adjusted for price changes. • The real GDP of a particular year is calculated by multiplying the amount of output of that year by the prices in the base year. • It is also called GDP in volume terms.

A change in nominal GDP does not necessarily imply a change in the real output of an economy.

- Relationship between real GDP and nominal GDP
  - Real GDP is not affected by changes in prices.
  - The relationship between real GDP and nominal GDP:

$$\text{Real GDP} = \text{Nominal GDP} \times \frac{\text{Price index of the base year}}{\text{Price index of the current year}}$$

  - **Implicit price deflator of GDP** is a measure of the change in prices of final goods and services included in the calculation of GDP.

$$\text{Implicit price deflator of GDP} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

  - When nominal GDP increases (decreases), real GDP may increase, decrease or remain unchanged, depending on the changes in the price level.

			Real GDP
Percentage increase in nominal GDP	>	Percentage increase in prices	increases
	<		decreases
	=		remains unchanged

			Real GDP
Percentage decrease in nominal GDP	>	Percentage decrease in prices	decreases
	<		increases
	=		remains unchanged

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16.2 Per Capita GDP

- Per capita GDP shows the average welfare of people in an economy.

$$\text{Per capita nominal GDP} = \frac{\text{Nominal GDP}}{\text{Population size}}$$

$$\text{Per capita real GDP} = \frac{\text{Real GDP}}{\text{Population size}}$$

A country with a higher per capita GDP than another country may not have a higher GDP, depending on the population sizes of the two countries.

16.3 Growth Rate of GDP

- Growth rate of a variable

$$\text{Growth rate} = \frac{x_1 - x_0}{x_0} \times 100\%$$

- Growth rate of four GDP measurements

$$\text{Growth rate of nominal GDP} = \frac{\text{Nominal GDP of the current year} - \text{Nominal GDP of the previous year}}{\text{Nominal GDP of the previous year}} \times 100\%$$

$$\text{Growth rate of real GDP} = \frac{\text{Real GDP of the current year} - \text{Real GDP of the previous year}}{\text{Real GDP of the previous year}} \times 100\%$$

$$\text{Growth rate of per capita nominal GDP} = \frac{\text{Per capita nominal GDP of the current year} - \text{Per capita nominal GDP of the previous year}}{\text{Per capita nominal GDP of the previous year}} \times 100\%$$

$$\text{Growth rate of per capita real GDP} = \frac{\text{Per capita real GDP of the current year} - \text{Per capita real GDP of the previous year}}{\text{Per capita real GDP of the previous year}} \times 100\%$$

- When the growth rate is positive (negative): There is an increase (a decrease) in economic activities.
- When the growth rate is zero: The level of economic activity is the same over the period.

When the growth rate of real GDP decreases but remains positive, real GDP still increases and people's living standard is improved.

16.4 Gross National Income (GNI)

- **Gross national income** (GNI) measures **the total income earned by residents of an economy from engaging in various economic activities during a specific period, irrespective of whether the economic activities are carried out within the economic territory or not**.
- Meaning of residents

For individuals	For organisations
residents refer to those who normally stay in the economic territory of an economy for at least 12 months, or intend to do so, irrespective of their nationalities	residents refer to those which ordinarily operate in the economic territory of an economy

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- Relationship between GDP and GNI
  - GNI may not have the same value as GDP.
  - **Net factor income from abroad**

$$\text{Net factor income from abroad} = \text{Factor income earned by residents outside the economic territory} - \text{Factor income earned by non-residents within the economic territory}$$

  - Calculating GNI from GDP

$$\text{GNI} = \text{GDP} + \text{Net factor income from abroad}$$

$$\text{GNI} = \text{GDP} + \text{Factor income earned by residents outside the economic territory} - \text{Factor income earned by non-residents within the economic territory}$$

GNI will be smaller than GDP if net factor income from abroad is negative.

16.5 Uses and Limitations of National Income Statistics

- Uses of national income statistics
  - Reflection of the standard of living
  - Assessing the economic performance of an economy
  - Formulation of government policies
  - Basis for international comparison
- Limitations of national income statistics
  - Effect of price fluctuation
  - Effect of population size
  - Effect of income distribution
  - Leisure time
  - Effect of composition of output
  - Unreported production activities
  - Environmental quality

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